



FACULTY OF SCIENCE AND TECHNOLOGY
LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT

PROGRAMME: BSSE/BSCS/BIT

YEAR/SEM: YEAR 2 SEM 1

COURSE CODE: BIT2115

PROBLEM NAME: EXPLICIT & IMPLICIT DATA CONVERSION

STUDENT DETAILS:

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REG NUMBER	25/1707/BSCS-S



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Section 1

Logic Building & Programming

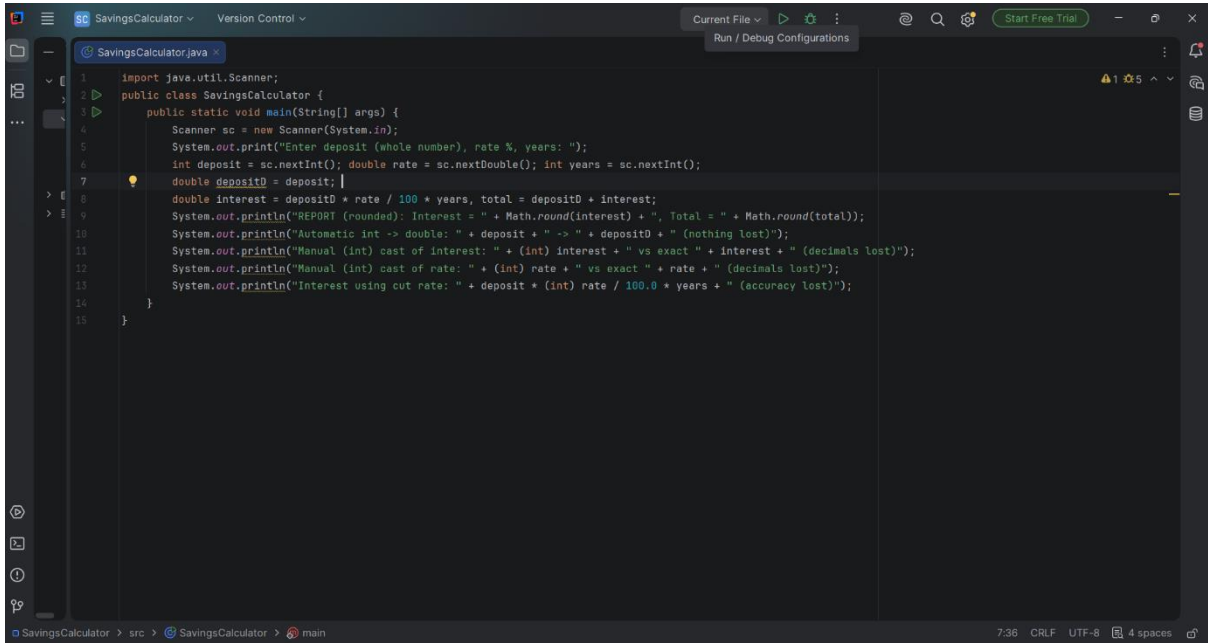
Lines of code & explanations

Line Number	Explanation
1	Imports the Scanner class so the program can read the user's input from the keyboard.
2	Declares the class SavingsInterestCalculator.
3	The main method, where the program starts running.
4	Creates a Scanner object (sc) that reads what the user types.
5	Displays a prompt asking the user to enter the deposit, the interest rate and the number of years.
6	Reads the deposit as a whole number (int), the rate as a decimal (double) and the years as a whole number (int).
7	IMPLICIT (automatic) conversion: the int deposit is widened to a double. This is safe because a double can hold every int value, so no data is lost.
8	Calculates the interest (deposit x rate / 100 x years), which produces a decimal value, and then the total (deposit + interest).
9	Prints the summary report for management. Math.round() rounds the interest and the total to whole numbers for accounting purposes.
10	Shows the int deposit and its double form (for example 1000 -> 1000.0), demonstrating that automatic conversion loses nothing.
11	EXPLICIT (manual) conversion: (int) interest casts the double to an int. The decimal part is chopped off, not rounded, so it is compared with the exact value to show the loss.
12	Casts the rate to an int. Decimals are lost, so a manual cast is needed and accuracy is affected.
13	Recalculates the interest using the cut-down rate, which gives a different (wrong) amount. This shows how data accuracy is affected during conversion. 100.0 keeps the division in decimals.
14	Closes the main method.
15	Closes the class.

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Detailed Screenshots and caption descriptions



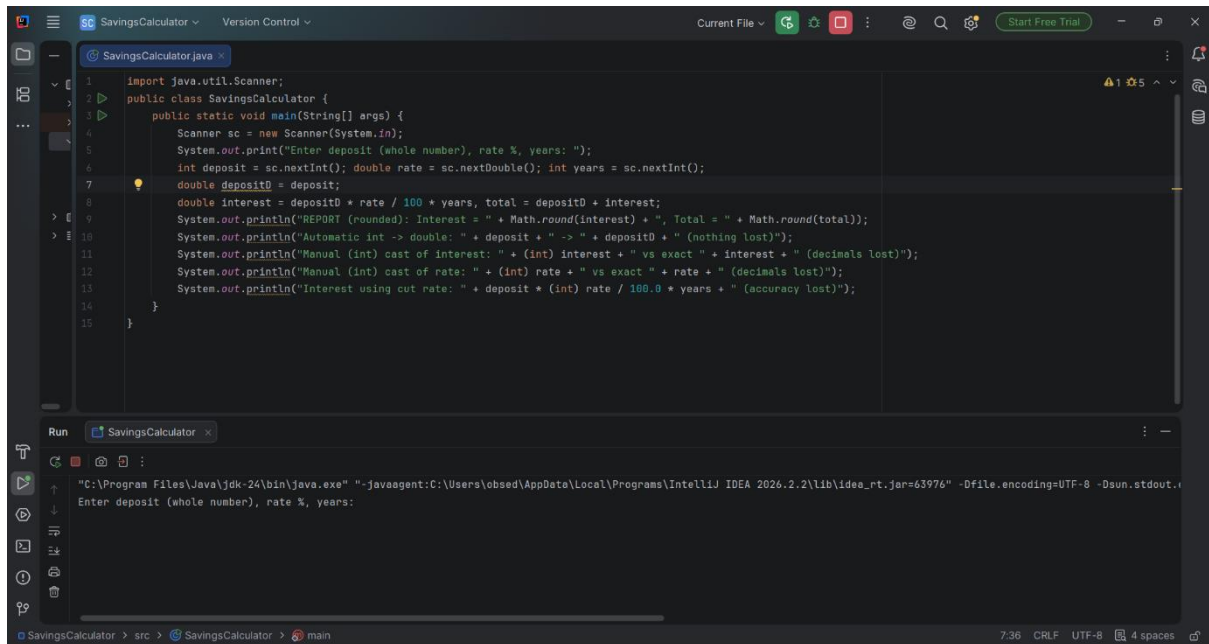
```
1 import java.util.Scanner;
2 public class SavingsCalculator {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter deposit (whole number), rate %, years: ");
6         int deposit = sc.nextInt(); double rate = sc.nextDouble(); int years = sc.nextInt();
7         double depositD = deposit;
8         double interest = depositD * rate / 100 * years, total = depositD + interest;
9         System.out.println("REPORT (rounded): Interest = " + Math.round(interest) + ", Total = " + Math.round(total));
10        System.out.println("Automatic int -> double: " + deposit + " -> " + depositD + " (nothing lost)");
11        System.out.println("Manual (int) cast of interest: " + (int) interest + " vs exact " + interest + " (decimals lost)");
12        System.out.println("Manual (int) cast of rate: " + (int) rate + " vs exact " + rate + " (decimals lost)");
13        System.out.println("Interest using cut rate: " + deposit * (int) rate / 100.0 * years + " (accuracy lost)");
14    }
15 }
```

Figure 1:Class Main showing project structure

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Output and caption descriptions



```
1 import java.util.Scanner;
2 public class SavingsCalculator {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.print("Enter deposit (whole number), rate %, years: ");
6         int deposit = sc.nextInt(); double rate = sc.nextDouble(); int years = sc.nextInt();
7         double depositD = deposit;
8         double interest = depositD * rate / 100 * years, total = depositD + interest;
9         System.out.println("REPORT (rounded): Interest = " + Math.round(interest) + ", Total = " + Math.round(total));
10        System.out.println("Automatic int -> double: " + deposit + " -> " + depositD + " (nothing lost)");
11        System.out.println("Manual (int) cast of interest: " + (int) interest + " vs exact " + interest + " (decimals lost)");
12        System.out.println("Manual (int) cast of rate: " + (int) rate + " vs exact " + rate + " (decimals lost)");
13        System.out.println("Interest using cut rate: " + deposit * (int) rate / 100.0 * years + " (accuracy lost)");
14    }
15 }
```

Run SavingsCalculator

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\obsed\AppData\Local\Programs\Intel\IJ IDEA 2026.2.2\lib\idea_rt.jar=63976" -Dfile.encoding=UTF-8 -Dsun.stdout.
Enter deposit (whole number), rate %, years:

Figure 2: Output after running the program

Output Explanation

When the program is executed, it displays the following, in order:

1. It asks the user to enter the deposit (whole number), the interest rate and the number of years.
2. The report line shows the interest and total rounded to whole numbers with `Math.round()`, as required for accounting.
3. The automatic conversion line shows the int deposit becoming a double. This is safe because nothing is lost.
4. The manual cast of the interest to int cuts off the decimals, so it differs from the exact value.
5. The manual cast of the rate to int also loses its decimals.
6. The last line shows the interest recalculated with the cut rate, which is wrong, so accuracy is lost.

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Conclusion: widening (int to double) is safe and automatic, while narrowing (double to int) needs a manual cast and loses accuracy.

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Section 2
Version Control

Repository Overview

Repository Name	
Repository URL	
Branches	

Commit #1 On Branch #1

Commit Summary

Commit Name	
Commit ID	
Commit Description	
Branch Affected	

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Section 3

What Have You Learned From The Task?

Skill Set	Description
Implicit conversion	Understood that Java automatically converts a smaller data type to a larger one, such as int to double, and that this is safe because no data is lost.
Explicit conversion (type casting)	Learned to use casts such as (int) to convert a larger type to a smaller one, and saw that decimals are chopped off, which reduces accuracy.
Rounding and reading input	Used Math.round() to produce rounded figures for reports and Scanner to accept user input in a Java program.